



Plotting Data from a Web Transcript

So now let us go back to understanding how to upload data from a website. We have done uploading data from a CSV. So in the live class we will show you how to load data directly from a CSV file and then do operations.

I just gave you a very basic introduction to do CSV. I urge you to go visit the examples directly from the documentation and develop and we will also do that in the live class. So now let us try to fetch and plot data from the web.

So these days especially if you are working in let us say in finance or you are tracking stock markets or you are working in climate, the data is hosted on a particular website which would then be imported into your particular domain and then you will be using them for doing the visualisations. So in this case we are trying to import the weather data of Seattle. So this is the URL.

So first you have to identify the URL from which you are trying to get the data from. Always remember whenever you are running you should have the necessary libraries in place before you do anything. So I am importing this particular URL.

So I know the URL. So you are defining a variable called URL is equal to so and so. So this is a CSV.

So you have imported the particular variable. Now what are you trying to do? You are then just using the you are creating a data frame. See Python needs a data frame for all operations.

So understand the logic why we are doing each of the step. So what again this is a CSV data. So we are saying using the PDF underscore I mean `pd.read` underscore CSV command and parsing the URL.

We are just parsing the URL which is the variable which indicates this particular place and we want dates to come from the particular date column. `Pars underscore dates` is equal to `date`. The date column from the website.

So let me just also try this to showcase how does this data look like. So this is how the data looks from the website. It has date, precipitation, temperature, maximum and wind and weather.

So the column is there and you are telling the Python environment to say you use this particular URL which captures which sends you to this particular website and you use past dates from the date column of that particular data. So you have done that then again you are using code. So now that the data is ready for you, you are trying to plot it.



So I am just going there and plotting the data. So what I am doing weather underscore set index the date. The elements are ordered in terms of the date column.

Then you are plotting the figure. You are setting the figure size. Temperature, weather, max.

What is the maximum temperature is your variable of interest and you are plotting with the colour crimson. So x label is date, y label is this. Once you have subsetting the data you are able to do the graphic.

So it's simply the point where you are trying to fix the date column as the index and then set the date index for easier times it is plotting. Why? Because you need this to be ordered on the date otherwise you will get random numbers in between random dates in between and it is not making any sense. So you needed it for ordering the data on terms of the date because date is the historical series that you are interested in and you are plotting the temperature maximum.

So you are just creating a subset of the plot data. This is called subsetting or calling out a particular column in Python where you are using df underscore weather right is the data frame that you have created and in that you are calling out the particular column called temperature max. Now what other columns are there? So let us see let us use this df underscore weather.

See this is the data frame and it says okay you want to look at the head. Let us do that part and this doing this run selection so you are getting this.

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